

Technical Document #295: Software Engineering - Comprehensive Overview

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Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

Continuous deployment (CD) automatically deploys code changes to production. It enables rapid, reliable releases.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Key Takeaways:

- Code review examines code changes before merging.
- Continuous deployment (CD) automatically deploys code changes to production.
- Technical debt represents the cost of choosing easy solutions over better ones.